



**Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam –
603 110**

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of CSE

Team Project Topics

Regulations – R2021

Degree & Branch	BE CSE			Semester	V
Subject Code & Name	UCS 2501 Computer Networks				
Academic Year	2023-2024 ODD	Batch	2021-2025	Due Date	11.10.2023

Implement the following problems which is allocated to your team using the knowledge you gained from CO3, CO4 & CO5 of the course. Prepare documentation for the same. You can use Network Simulator, Cisco Packet Tracer, Java, Python, C and Wireshark tool as demanded by the project requirement.

Marking Scheme:

	Marks
Implementation	25
Upload in your Github page	05
Documentation with ReadMe file and Github link	10
Viva	10
Total	50

Submission Due Date: 13/11/2023 (upload in LMS)

Documentation Format: (Minimum 15 pages)

Title

Team members regno, name

Index

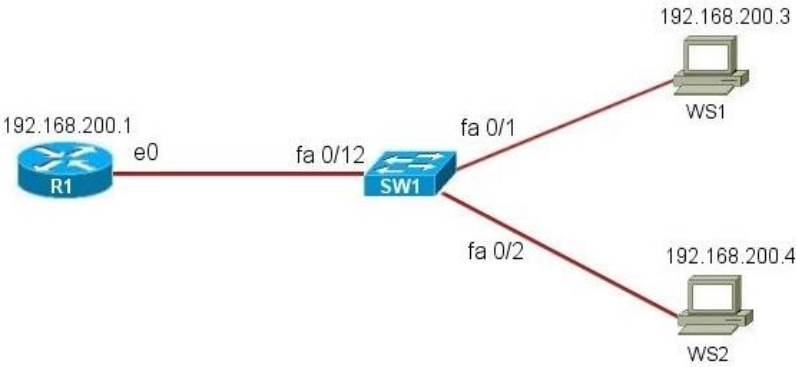
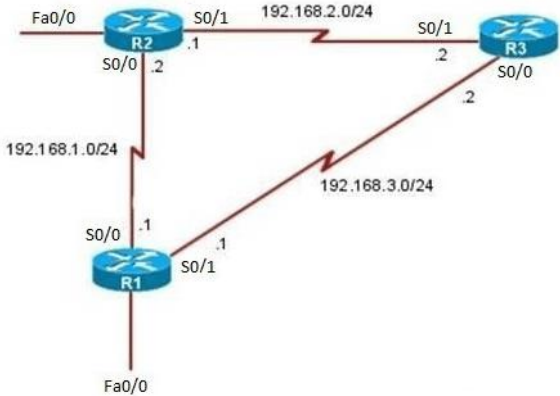
Contents – Problem Definition, Protocol/ Method explanation, Topology, Code, Output, Learning

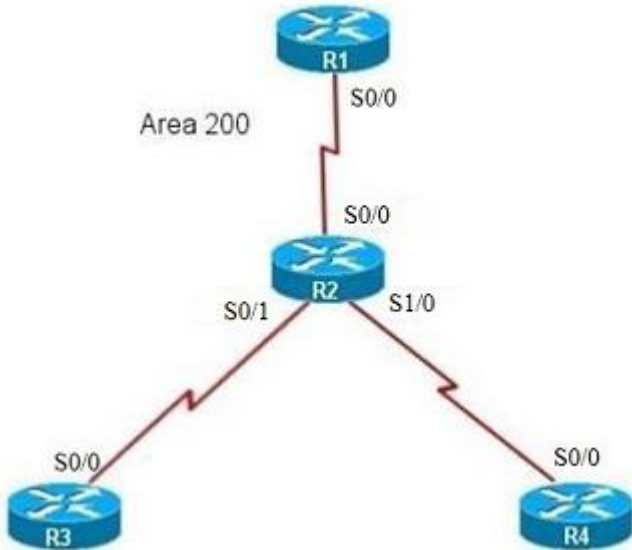
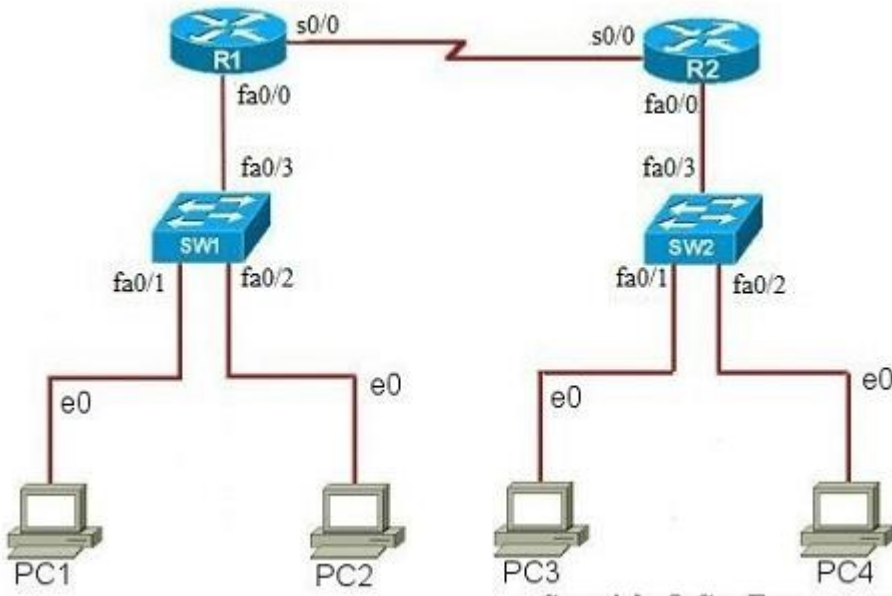
Outcomes

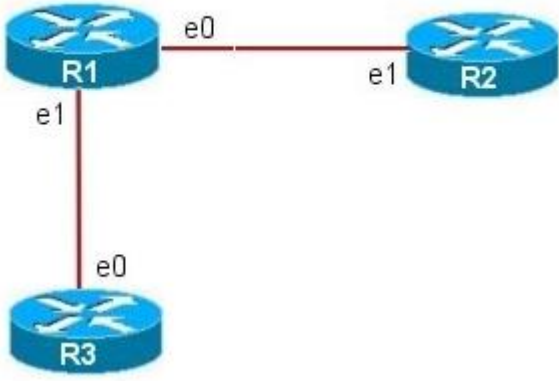
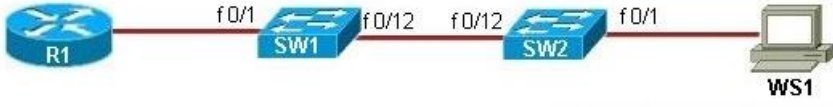
Readme File

Github Link

Useful Link : <https://routersimulator.certexams.com/router-labs/index.html>

1	Build the following network, implement VLAN and analyse it. https://routersimulator.certexams.com/router-labs/configuring-devices-to-support-vlans.html 	(CO3, K4)
2	Design and build the following network, implement EIGRP and analyse it. https://routersimulator.certexams.com/router-labs/eigrp-routing-configuration-scenario.html 	(CO3, K4)
3	Develop and build the following network to implement RIP. Analyse the performance of it. https://routersimulator.certexams.com/router-labs/ripv2-routes.html 	(CO3, K4)
4	Build the following network and implement OSPF. Analyze the performance of it.	(CO4, K4)

	https://routersimulator.certexams.com/router-labs/ospf-troubleshooting-lab-scenario-1.html 	
5	Build the following network and implement NAT https://routersimulator.certexams.com/router-labs/nat-scenario-1.html 	(CO4, K3)
6	Build the following network and implement DHCP. https://routersimulator.certexams.com/router-labs/configuring-two-dhcp-clients-and-verification-commands.html	(CO5, K3)

	DHCP Server 	
7	Evaluate the performance of the network using Wireshark 	(CO5, K5)
8	Using NS2, evaluate the throughput performance of TCP Westwood, TCP New Reno. Plot the congestion window.	(CO4,K4)
9	Using NS2, evaluate the goodput performance of TCP Vegas, TCP New Reno. Plot the congestion window.	(CO4,K4)
10	Implement Selective Repeat ARQ using any programming language.	(CO4,K4)
11	Implementation of finding the first and last fragment of TCP using python.	(CO4,K4)
12	Simulation of allocating segment number based on the number of bytes transmitted and its acknowledgment number	(CO4,K4)
13	Make your home PC as a server and use FTP to retrieve files from this server to any host on the internet.	(CO5,K5)
14	Simulation of TCP Congestion Window increase of Tahoe using socket programming in C.	(CO4,K4)
15	Simulation of RIP Table by exchanging the updates with its neighbour alone using Java / C programming	(CO4,K4)
16	Implementation of NAT using C.	CO5,K3
17	Implementation of address aggregation concept in C	CO5,K3

Course Incharge

Dr. S. V. Jansi Rani & Dr. N. Sujaudeen